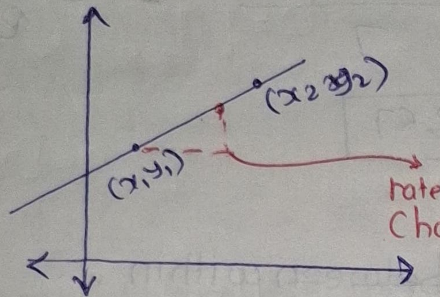
* Differential Calculus.

Slope :- :- Rate of change of ~~of~~ one variable with respect to another.

:- Slope indicate the ratio of the vertical change (rise) to horizontal change (run) between two points on a line.

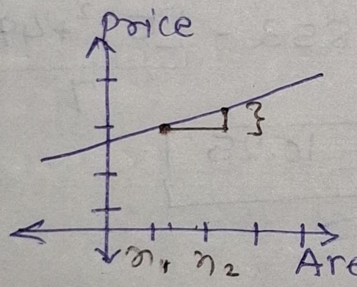


$$\text{slope} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x} = \frac{\text{Rise}}{\text{Run}}$$

where $y_2 - y_1 \longrightarrow$ vertical Rise
$x_2 - x_1 \longrightarrow$ Horizontal Rise

ex

Area	Price



Interpretation of slope :-

1] +ve slop :- slope > 0
:- line rises as it moves from left to right, larger the slope → steeper the line.

2] -ve slope :- slope < 0
:- line falls as it moves ~~fe~~from left to right. the ~~steeper~~ more -ve the ~~line~~ slope, the steeper the line in -ve direction.

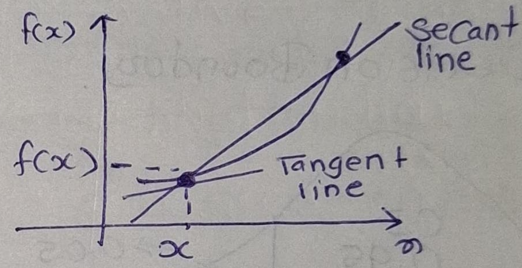
3] Zero slope :- ~~st~~ line is Horizontal. No vertical change.

4] undefined slope
$x_2 = x_1$
line is vertical.

slope equation of line

$y = mx + c$ → ~~y~~ c is y intercept where line crosses y axis
↓
m is slope

* Derivative

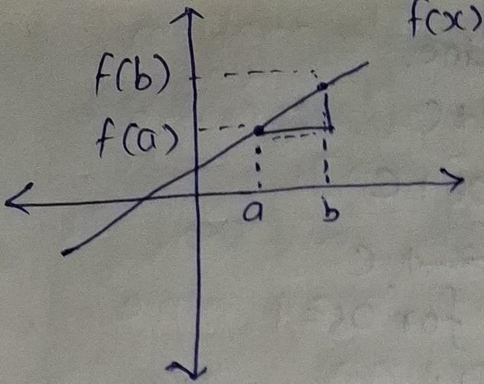


$\text{slope} = \frac{\partial y}{\partial x}$
instantaeous rate of change.

Mathematical Notation of Derivative with limits

Derivative : represents the rate at which is changing at given point.
it is essentially the slope of tangent line to the function graph at point.

$f(x) = mx + c$

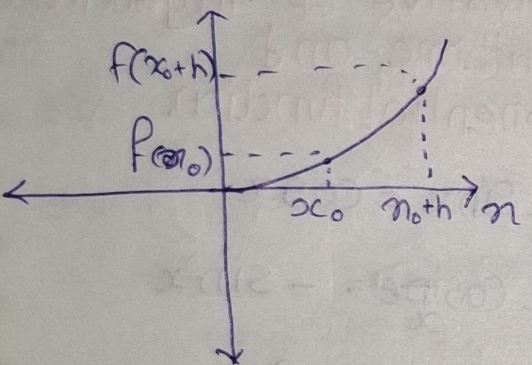


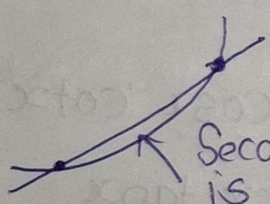
$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{\Delta y}{\Delta x}$$

$$= \frac{y_2 - y_1}{x_2 - x_1} = \frac{f(b) - f(a)}{b - a}$$

$$\frac{\Delta y}{\Delta x} = \frac{f(b) - f(a)}{b - a}$$

Now, if we have Curve





Secantline slope is going to give the average rate of change.

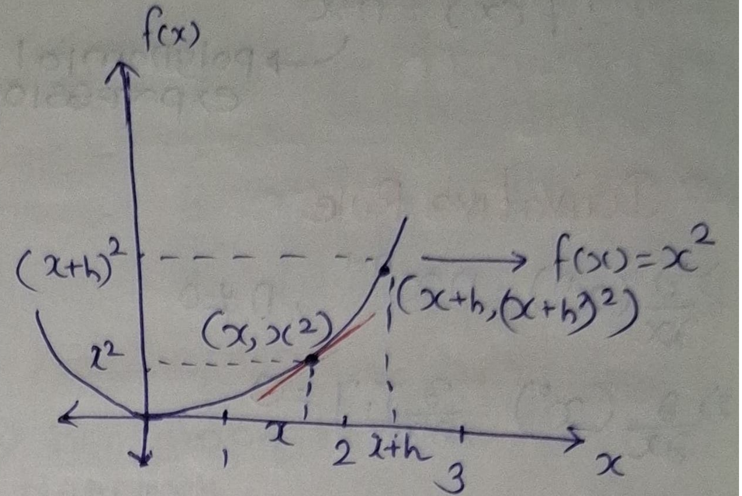
$$\text{Slope} = \frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{f(x_0+h) - f(x_0)}{(x_0+h) - x_0}$$

$$\text{Slope of Secant line} = \frac{f(x_0+h) - f(x_0)}{h}$$

$$f'(x) = \lim_{h \to 0} \frac{f(x_0+h) - f(x_0)}{h}$$

$$f'(x) = \frac{\partial y}{\partial x} = \frac{\partial (f(x))}{\partial x}.$$

• finding Derivate at a point.



$$f'(x) = \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$$

$$f'(x) = \lim_{h \to 0} \frac{\cancel{x^2} + 2xh + h^2 - \cancel{x^2}}{h}$$

$$\lim_{h \to 0} \frac{(2x + h)\cancel{h}}{\cancel{h}}$$

$$f(x) = \lim_{h \to 0} 2x + h$$

$$f(x) = 2x$$

it means, Derivative of x^2 is $2x$

$$\boxed{x^n = nx^{n-1}}$$

Mathematically

$$\therefore \frac{\partial (f(x))}{\partial x} = \frac{\partial (x^2)}{\partial x} = 2x.$$

* Power Rules in Derivative

$f(x) = x^n$ where $n \neq 0$

$\therefore f'(x) = n x^{n-1}$ $\longrightarrow$ polynomial expression

* Derivative Rule

1) $\frac{\partial}{\partial x}(x^n) = n x^{n-1}$, $n \neq 0$

2) $\frac{\partial}{\partial x}(x^0) = \frac{\partial}{\partial x}[1] = 0$

3) $\frac{\partial}{\partial x}[c] = 0$

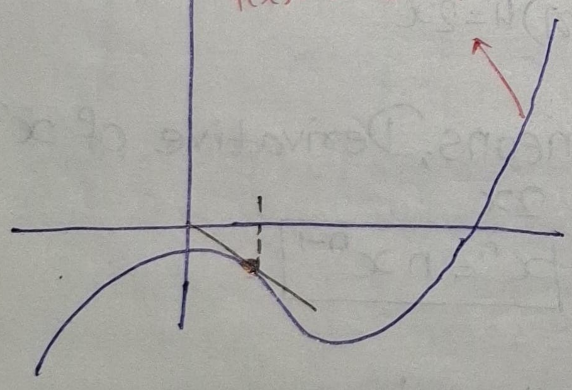
4) Sum of two functn.

$f(x), g(x)$

$\frac{\partial}{\partial x}[f(x) + g(x)]$

$= \frac{\partial}{\partial x} f(x) + \frac{\partial}{\partial x} g(x)$

* Equation of tangent of polynomial.



$f(x) = x^3 - 6x^2 + x - 7$

$f(1) = 1 - 6 + 1 - 7 = -11$

Now

$f'(x) = \frac{\partial}{\partial x}(x^3 - 6x^2 + x - 7)$

$= 2x^2 - 12x + 1 - 0$

$f'(x) = 2x^2 - 12x + 1$

$f'(x) = 2(1) - 12 + 1 = -9$

tangent line.

$y = mx + c$

$y = -9x + c$

$\cancel{-11 = -9x + c}$

$\cancel{-11 = -9}$ for $x = 1$

$y = -11$

$-11 = -9 + c$

$c = -11 + 9$

$c = -2$

$\therefore$ equation is

$y = mx + c$

$y = -9x - 2$

* Derivative of trigonometric logarithmic and exponential function

1) $\frac{\partial}{\partial x} \sin x = \cos x$

2) $\frac{\partial}{\partial x} \cos x = -\sin x$

3) $\frac{\partial}{\partial x} \tan x = \sec^2 x$

4) $\frac{\partial}{\partial x} \operatorname{cosec} x = -\cos x \cot x$

5) $\frac{\partial}{\partial x} \sec x = \sec x . \tan x$

6) $\frac{\partial}{\partial x} \cot x = -\operatorname{cosec}^2 x$

7) $\frac{\partial}{\partial x}(e^x) = e^x$

8) $\frac{\partial}{\partial x}(a^x) = a^x \ln(a)$

9) $\frac{\partial}{\partial x}(\ln(|x|)) = 1/x$

10) $\frac{\partial}{\partial x}(\log_a x) = \frac{1}{x \ln(a)}$

* Product Rule.

$$\frac{\partial}{\partial x}\left[f(x).g(x)\right]$$

$$= f'(x)\, g(x) + \cancel{gx}\, f(x)\, g'(x)$$

* Division Rule

$$\frac{\partial}{\partial x}\left[\frac{f(x)}{g(x)}\right] = \frac{f'(x)\,g(x) - f(x)\,g'(x)}{[g(x)]^2}$$

* Chain Rule.

It is used to find the derivative of composite function, when a function is composed of other function the chain rule allow us to differentiate it with respect to the ~~int~~ inner most variable.

$$y = f(g(x))$$

$\therefore$ where $y = f(u)$

and $u = g(x)$.

$$\frac{\partial y}{\partial x} = \frac{\partial f}{\partial u} \times \frac{\partial u}{\partial x}$$

ex. $y = \sqrt{\sin(3x)}$

$$y = \frac{\partial}{\partial x}\sqrt{x} =$$

$f(u) = \sqrt{u}$ $\quad u = \sin 3x$.

$\therefore y = u^{1/2}$

$g(v) = \sin(v)$ & $V = 3x$

$v = h(x) = 3x$.

$$y = \frac{1}{2}(\sin(3x))^{\frac{1}{2}-1} \cdot \cos(3x)$$

$$\cdot\ 3$$

$$= \frac{3}{2}\left[\sin(3x)\right]^{-1/2} \cdot \cos(3x)$$

$$= \frac{3}{2}\,\frac{\cos(3x)}{\sqrt{\sin(3x)}}$$

* Application of chain Rule.

1) Backpropogation In Neural Network

$\therefore$ Training Deep Learning Model.

chain rule for calculating the gradient of loss function with respect to the weights that are initialized.

2) Gradient Descent Optimization.

$\therefore$ Linear Regression

$\rightarrow$ goal is to minimize a cost function

$\therefore$ update slopes.

$\rightarrow$ Gradient Descent optimizer

3) Feature Enginerring

4) Regularization technique.

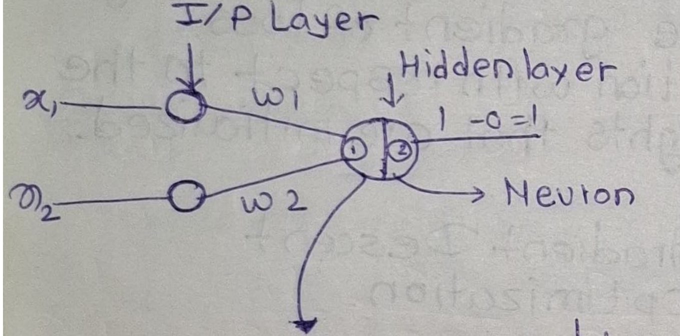
5)

* Perceptron. {Binary classifier}
[Artificial Neuron]

① Input layer
② Hidden layer
③ Weights
④ Activation function.

ex. Dataset.

IQ	No. of Study hrs	O/P
95	3	0
100	4	1
150	5	1



bias.

① ∴ $\Sigma x_i w_i + b$

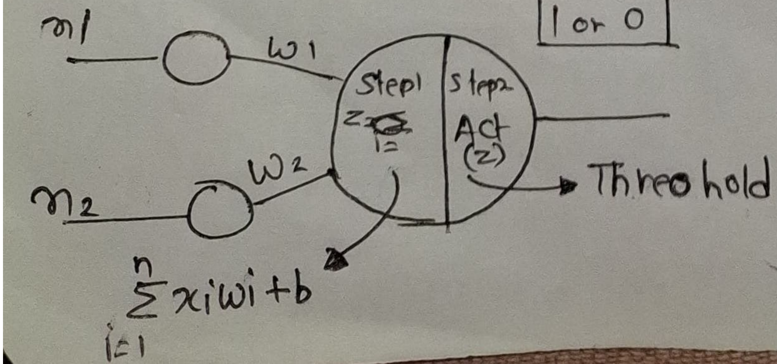
$z = w_1 x_1 + w_2 x_2 + b$

~~b=~~

② Activation (z)

∴ activation funcⁿ Transform the output betⁿ 0 to 1
-1 to 1

Advantage and disadvantage of perceptron model



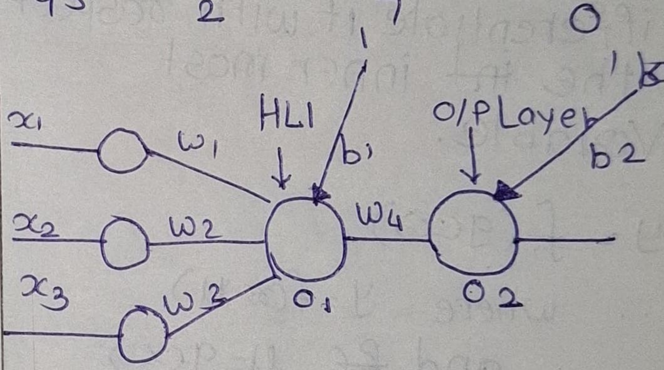
• Linear Separable problem

• MultiLayer Perceptron model [ANN]

1) forward propogation
2) Backward propogative
3) Loss function
4) optimizers
5) Activation function.

ex.

IQ	Study Hrs	Play hours	O/P
95	4	4	1
100	5	2	1
95	2	7	0



① forward propogation

Step1 $z = x_1 w_1 + x_2 w_2 + x_3 w_3 + b_1$

$z = \sum_{i=1}^{n} w_i^T x_i + b$

Step2 activation (z)

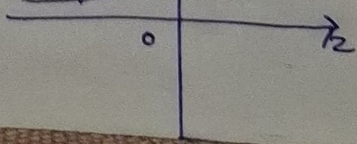
ex:-
take $\begin{array}{ccc} w_1 & w_2 & w_3 \\ \end{array}$ [0.01, 0.02, 0.03]

bias [0.001]

$z = 95 \times 0.01 + 4 \times 0.02 + 4 \times 0.03 + 1 \times 0.01$

$= 1.151$

Step 2: Activation(z)
sigmoid



$$\text{Sigmoid} = \frac{1}{1+e^{-z}}$$

Hidden layer

$$f(z) = \frac{1}{1+e^{-1.151}} = 0.759$$

$O_1 = 0.759$

Hidden layer 2

$w_4 = 0.02$
$b_2 = 0.03$

$$z = O_1 * w_1 + b_2$$
$$= 0.759 \times 0.02 + 0.03$$
$$= 0.04518$$

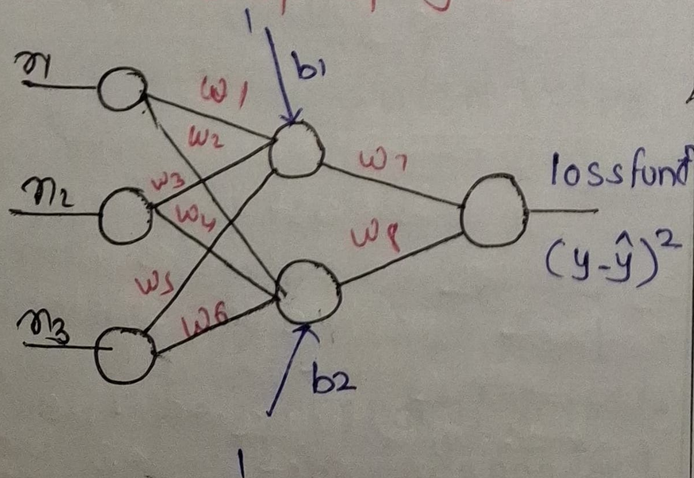
Activation.

$$\frac{1}{1+e^{-(0.04518)}} = 0.51129$$

$\therefore$ loss function $= (1 - 0.51129)$
≈ 0.49
↓
~~oel~~ Error.

Aim :- to reduce / minimize the error.
↓
using Backword propogation.

Backpropogation.

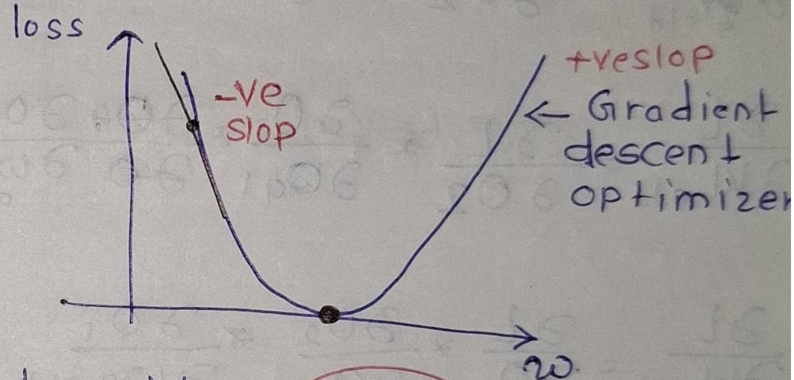


$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

weight - updation Formula.

$$w_{7new} = w_{7old} - \eta \frac{\partial L}{\partial w_{7old}}$$

$$\boxed{W_{new} = W_{old} - \eta \frac{\partial L}{\partial W_{old}}}$$



$W_{new} = W_{old} - \eta\,(-ve)$ — learning rate
$= W_{old} + \eta\,(+ve)$

$W_{new} >> W_{old}$

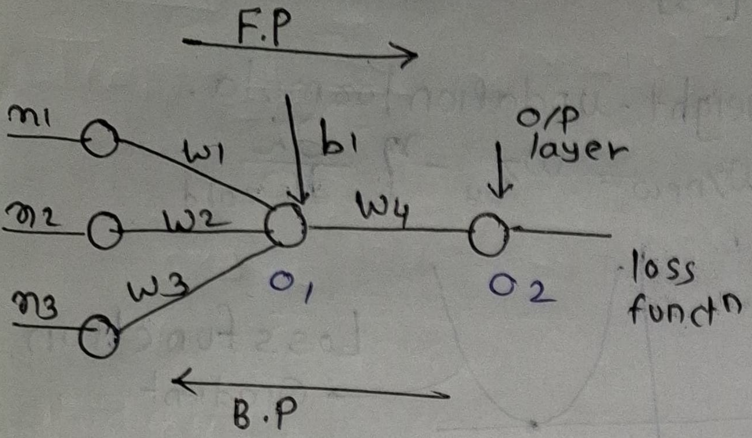
η - value start with minimum value.

$W_{new} = W_{old} - \eta\,(+ve)$
$= W_{old} - \eta\,(+ve$

$W_{new} << W_{old}$

When 'W' reaches globalminima
$\therefore W_{new} = W_{old}$

use of chain rule in backpropogation.



$$w_{new} = w_{old} - \eta \frac{\partial L}{\partial w_{old}}$$

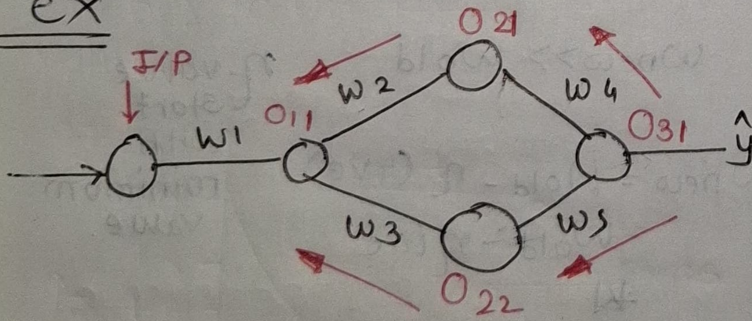
$\therefore$

$$\frac{\partial L}{\partial w_{4old}} = \frac{\partial L}{\partial O_2} * \frac{\partial O_2}{\partial w_4}$$

$$\frac{\partial L}{\partial w_{1old}} = \frac{\partial L}{\partial O_2} * \frac{\partial O_2}{\partial O_1} * \cancel{\frac{\partial O_4}{\partial O}} \frac{\partial O_1}{\partial w_{1old}}$$

$$\frac{\partial L}{\partial w_{2old}} = \frac{\partial L}{\partial O_2} * \frac{\partial O_2}{\partial O_1} * \frac{\partial O_1}{\partial w_{2old}}.$$

ex



$$W_{new} = W_{old} - \eta \frac{\partial L}{\partial W_{old}}$$

$$\frac{\partial L}{\partial W_{1old}} = \left[\frac{\partial L}{\partial O_{31}} * \frac{\partial O_{31}}{\partial O_{21}} * \frac{\partial O_{21}}{\partial O_{11}} * \frac{\partial O_{11}}{\partial w_{1old}}\right]$$

$$+$$

$$\left[\frac{\partial L}{O_{31}} + \frac{\partial O_{31}}{\partial O_{22}} + \frac{\partial O_{22}}{\partial O_{12}} + \frac{\partial O_{12}}{\partial w_{1old}}\right]$$